

Plausible Mechanisms for Hallucination in Generative AI Systems

A Response-Production Framework

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Working Paper Version 1.0
July 2026

Abstract

Hallucination in generative AI systems is commonly identified when a representational claim is contradicted by, or lacks support required from, an applicable source of truth. Although hallucination is widely measured and discussed, the processes through which it arises remain incompletely understood. This paper develops a response-production framework for examining plausible sources of hallucination. It traces response production from model formation, through autoregressive generation and decoding, to retrieval, context construction, and other deployment processes, identifying potential loci of error generation across these stages. The framework also shows how processes operating at different stages may interact. Existing evidence supports various elements of the proposed pathways to different degrees. The framework suggests a response-production approach to hallucination evaluation and management and generates testable hypotheses for future research.

Keywords: hallucination; response production; model approximation; autoregressive realization; deployment mediation; retrieval-augmented generation

A. Introduction

Hallucination is widely recognized as a limitation of generative AI. Research has developed definitions, benchmarks, detection methods, and mitigation strategies across natural-language-generation tasks ([1], [2]). Much of this work asks whether a response contains an incorrect or unsupported claim and how often such claims occur. This paper begins after that determination has been made. Its question is: where in response production may an identified hallucination have come from?

The question matters because the trained model is only one part of the system that produces a response. A model is formed from selected historical data through particular objectives, architectures, capacities, and optimization procedures. At inference, it generates conditional predictions that are converted into one response through a decoding procedure. In deployment, retrieval, prior interactions, tools, system instructions, context construction, and post-generation processes can change both the information presented to the model and the response ultimately delivered to the user.

Existing research provides important evidence about each of these elements. Prior surveys organize hallucination by task, output type, cause, detection method, and mitigation strategy ([1], [2]). More focused studies examine temporal and knowledge limitations, predictive uncertainty, decoding, response length, error propagation, retrieval augmentation, and long-context use. This paper builds on that work by locating these findings within the sequence that produces a delivered response. The purpose is not to offer

another exhaustive taxonomy, but to connect otherwise dispersed findings to three production loci and to show how conditions at one locus can alter the effects of another.

This paper organizes plausible sources of hallucination into three mechanism families. *Model Approximation* concerns limitations associated with the formation of a finite predictive model from bounded data through design-contingent choices. *Autoregressive Realization* concerns the process through which conditional predictions are decoded into a particular response path and earlier selections condition what follows. *Deployment Mediation* concerns retrieval, context construction, tools, interaction history, and post-generation processes that shape what the model receives and what the user sees.

The families identify principal loci of susceptibility; they are not mutually exclusive. A limitation inherited from model formation may be corrected by retrieval, amplified by poor context construction, realized through decoding, and extended through later generation. Conversely, deployment and realization controls may prevent a model limitation from appearing in the delivered response. The same type of observed error can therefore arise through different production paths.

The contribution is therefore an organizing framework rather than a complete causal theory. It supports a response-production approach to evaluating and managing hallucination and generates testable implications from the proposed structure. Section B presents the production sequence. Sections C through E examine the three mechanism families. Section F explains their interaction. Section G discusses interpretation, management, evidence, limitations, and conclusions.

B. From Model Formation to Delivered Response

The framework follows one interaction at time t from the trained model and deployment inputs to the response delivered to the user. Figure 1 summarizes the sequence, identifies the principal locus of each mechanism family, and shows why the three cannot be treated as independent compartments.

The sequence represents one response-production cycle. In systems that use repeated tool calls, test-time computation, search, or iterative revision, the sequence may recur: the output of one cycle becomes part of the operative context for the next. These systems add production paths and opportunities for interaction, but they do not necessarily require a separate mechanism family.

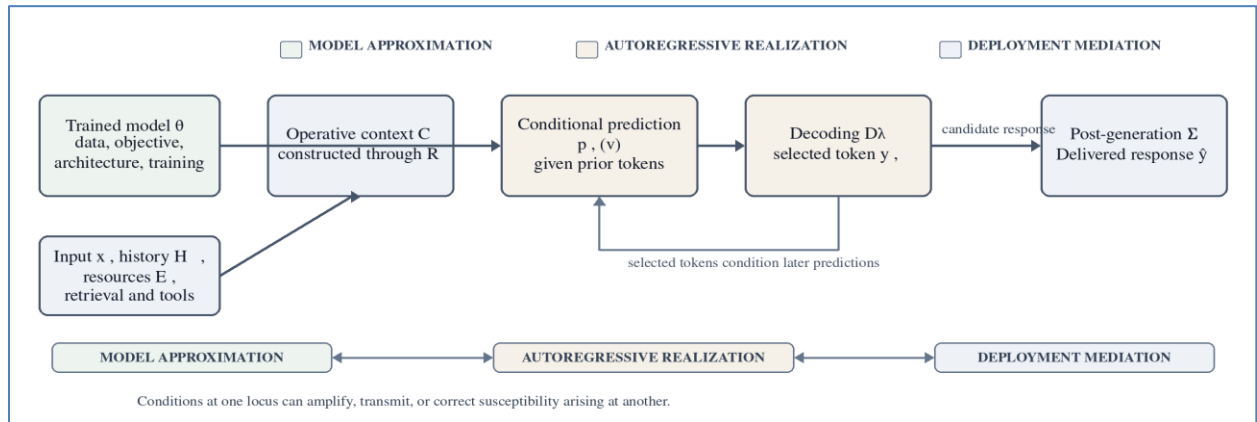


Figure 1. Response production and the three mechanism families. Each family identifies a principal locus of susceptibility; their effects may interact across the production sequence.

B.1 The trained model

Let θ_{t_0} denote the trained model used at response time t , where t_0 represents its effective temporal information boundary. The parameters embodied in θ_{t_0} were produced through selected data, a learning objective, architecture, capacity, optimization, and post-training completed before the later interaction.

Many contemporary large language models use decoder-only Transformer architectures derived from the attention-based structure introduced by Vaswani et al. ([3]). Text is divided into tokens and processed through repeated layers containing causal self-attention and feed-forward transformations. Pretraining adjusts the model parameters to predict subsequent tokens in large text corpora; post-training further shapes instruction following and response behavior. For the present analysis, these technical processes are summarized as the formation of a learned conditional predictor, θ_{t_0} .

B.2 Operative context

Let x_t denote the current input, H_{t-1} the prior interaction history, and E_t external resources potentially accessible at time t . A context-construction process R^L , operating under an effective constraint L , produces the operative context:

$$C_t = R^L(x_t, H_{t-1}, E_t). \quad (1)$$

The operative context C_t is the supplementary information actually presented for generation. The construction process may retrieve, select, omit, compress, order, or label information. Equation (1) therefore separates information potentially accessible to the deployed system from information actually supplied to the model.

B.3 Conditional prediction and realization

At token position k , the trained model produces a conditional distribution over the vocabulary V :

$$p_{t,k}(v) = P_{\theta_{t_0}}(y_{t,k} = v \mid x_t, C_t, y_{t,1:k-1}), \quad v \in V. \quad (2)$$

The distribution depends on the model, the current task, the operative context, and the response path already generated. A decoding procedure D_λ , operating under settings λ , selects the next token:

$$y_{t,k} \sim D_\lambda(p_{t,k}). \quad (3)$$

Repeating this operation produces a candidate response $\tilde{y}_t$. Because every selected token enters the conditioning history in Equation (2), the response path depends on earlier selections. A deployed system may then apply a post-generation process Σ :

$$\hat{y}_t = \Sigma(\tilde{y}_t). \quad (4)$$

where $\hat{y}_t$ is the response delivered to the user. Equations (1)–(4) provide notation for distinguishing the production stages; they are not a causal model of hallucination. Model Approximation principally concerns θ_{t_0} ; Autoregressive Realization concerns the conditional distribution, decoding, and generated path; and Deployment Mediation principally concerns R^L , external resources, interaction history, tools, and Σ .

C. Model Approximation

A generative AI model is not a truth set, a database, or a lossless representation of its training corpus. It is a finite predictive model produced from selected observations through a particular learning process.

Model Approximation denotes the mechanism family through which bounded information and design-contingent model formation create susceptibility to factual error.

C.1 Bounded information

The clearest boundary is temporal. A model completed at an effective information boundary t_0 cannot acquire through that original training process facts that arise later. If an officeholder changes, a company reports new earnings, or a stock price moves after t_0 , the later information is absent unless it enters through a subsequent update or deployment process. Current-knowledge studies find that models struggle with changing information and that their operative temporal knowledge can be older than a nominal training cutoff ([4], [5]).

The limitation is broader than time. Relevant information may have existed before t_0 but been absent from the selected corpus, private, obscure, task-specific, incorrectly represented, or too infrequent to exert reliable influence. A model may therefore lack a relationship even when that relationship was knowable before training ended.

The information boundary creates susceptibility, not a predetermined answer. An unaugmented model may produce an outdated claim, qualify its answer, state that current information is needed, or abstain. Deployment can also supplement the missing information. The boundary explains why the trained model alone is not adequate for every later task.

C.2 Predictive objective and factual adequacy

The standard language-model objective rewards prediction of tokens found in text. It does not directly compare every proposition implied by generated text with an authoritative truth set. Textual probability and factual truth overlap because true relationships appear in language, but they are not identical. A widely repeated misconception may be predictively strong, while a true but rare fact may exert weak influence. TruthfulQA provides evidence that models can reproduce common human falsehoods and that scaling alone does not necessarily improve truthfulness on such questions ([6]).

Post-training can modify these tendencies by further shaping how a model responds. Even so, a text-prediction objective does not guarantee that the most probable continuation will be factually correct.

C.3 Capacity, competing associations, and design contingency

Language models use a finite parameterization to encode patterns from a large and heterogeneous corpus. Research finds that factual storage and retrieval vary with model scale, training duration, data quality, fact frequency, compatibility among facts, and training order ([7], [8]). Work on knowledge overshadowing further suggests that a dominant association can obscure less prominent knowledge during generation ([9]).

Architecture and other design choices influence how information is represented and recovered. Their empirical success does not imply that they preserve every qualification, temporal index, source identity,

or factual relationship required by later tasks. Nor does an error in a Transformer establish that attention itself caused the error. The more limited conclusion is that the trained model is a design-contingent approximation. Its susceptibility to factual error can therefore change with the data, objectives, architecture, capacity, and post-training used to produce it.

D. Autoregressive Realization

A trained model and an operative context determine a conditional prediction for the next token, not a completed response. The response is realized sequentially: a decoding procedure selects a token from each conditional distribution, and that selection becomes part of the conditioning information for the next prediction. **Autoregressive Realization** refers to this process through which decoding converts successive conditional predictions into one response path.

D.1 Predictive uncertainty and decoding

The distribution in Equation (2) does not state the probability that a complete claim is true. It ranks possible next tokens according to the trained model and the conditioning information available at that point. A concentrated distribution can favor a false continuation. A more diffuse distribution may instead reflect several acceptable ways to express the same proposition. Predictive confidence and factual correctness are therefore related only imperfectly.

Uncertainty can nevertheless provide useful evidence about factual-error risk. Research on semantic entropy finds that variation across semantically distinct generations can help predict some hallucinations ([10]). Uncertainty is therefore a risk signal, not a determination that a response is wrong or an explanation of how an error arose.

Decoding determines how the system moves from a conditional distribution to a selected token. Greedy decoding selects the highest-ranked option. Sampling procedures may instead modify or restrict the distribution before making a selection. A deterministic procedure removes sampling variability, but it cannot make a falsely ranked continuation true. Conversely, inference-time decoding interventions have improved factuality without changing model parameters ([11]). These results show that realized error rates depend partly on how conditional predictions are converted into a response.

D.2 Response length

Response length can affect hallucination in at least two different ways. First, a longer response normally contains more factual assertions and therefore more opportunities for at least one error. This is **cumulative exposure**. It can increase the probability of a response-level error even when later assertions are no less reliable than earlier ones.

Second, a model may produce its most strongly represented facts first and rely on weaker or more specific associations when asked for additional detail. This is **facts exhaustion**. Controlled research on long-form factuality found this process to be the principal explanation for declining factuality in the settings examined ([12]). The finding need not apply uniformly across tasks. It nevertheless shows why the effect of response length should not be reduced to a single idea of uncertainty accumulation, error propagation, or long-context failure. The distinction also affects management. Concise response requirements can

reduce cumulative exposure, while requests for extensive factual detail may require stronger retrieval or verification when facts exhaustion is a concern.

D.3 Path dependence and propagation

Each selected token enters the history $y_{t,1:k-1}$ used to produce the next prediction. If a response first selects the wrong person, date, place, or premise, later generation may elaborate on that selection. The resulting text can be coherent with the earlier response while remaining false relative to the applicable source of truth.

Experiments on hallucination snowballing show that models can generate false explanations after an incorrect initial answer, including claims they can reject when those claims are evaluated separately ([13]). This evidence supports a propagation pathway: an early erroneous selection changes the conditioning information for what follows and can produce dependent errors. It does not imply that every long response deteriorates or that every later error results from an earlier one.

E. Deployment Mediation

The trained model is only one component of a deployed generative AI system. **Deployment Mediation** concerns the retrieval, tools, interaction history, context construction, and post-generation processes that shape what the model receives and what the user ultimately sees.

E.1 Information supplementation

Retrieval-augmented generation combines learned model representations with information retrieved from external sources ([14]). Retrieval can supply facts that arose after t_0 , private organizational information, designated documents, current project state, or specialized evidence unavailable through training. Tools can similarly provide calculations, database results, observations, or current market information at response time.

These capabilities allow a static model to remain useful for tasks that extend beyond its original information boundary. They do not update the model parameters. Instead, they supplement the information on which the response is generated. Deployment Mediation is therefore especially important for tasks that depend on current, private, or task-specific information.

E.2 Retrieval and context construction

Information does not influence generation merely because it exists in an accessible resource. Retrieval may omit the relevant source, return an outdated document, or select a textually similar but inapplicable passage. Even correctly retrieved material may be truncated, compressed, reordered, or stripped of temporal and source qualifications before it enters C_t .

Successful retrieval therefore differs from successful context construction. Retrieval makes information available for possible use. Context construction determines what is actually presented to the model and in what form. Evidence quantity and ordering have affected correctness in current-knowledge question answering ([5]). Providing retrieval capability alone is therefore not enough; its value depends on how the retrieved material is selected and presented.

E.3 Presence and effective use

Correct information can enter C_t without governing the response. The model may rely on a stronger learned association, fail to connect the evidence to the question, or blend conflicting sources. Research on long contexts also finds that performance can vary with the position of relevant information, including lower performance when decisive evidence appears in the middle ([15]).

This problem has two parts. Selecting, ordering, labeling, and placing evidence are deployment decisions. The system’s sensitivity to those decisions reflects the trained model and the realization of the response. Suppose decisive evidence is placed where the model uses it poorly and the response instead follows a stronger learned association. The resulting error is consistent with an interaction between Deployment Mediation and the other mechanism families. This distinction separates deployment choices from model behavior while allowing their interaction to explain the delivered result.

E.4 Tools and post-generation processes

Tools add further operations to response production. An error may arise because the system selects an unsuitable tool, submits an inadequate query, receives an erroneous result, or incorporates a correct result incorrectly. For a current stock-price question, a credible market-data service or verified financial-data API is generally more appropriate than asking a static language model to recall the answer.

The candidate response may also be verified, filtered, shortened, translated, or rewritten before delivery. These operations can remove an error. They can also introduce one or remove a necessary qualification. Deployment Mediation therefore affects both the information used during generation and the form of the response delivered to the user.

F. Interaction Across Mechanism Families

The three mechanism families describe different parts of one response-production sequence. The distinctions help locate potential sources of error, but they do not imply that the families operate independently. A condition or intervention at one production locus may change what happens at another.

Suppose a user asks who currently holds an office that changed after t_0 . The former officeholder may remain a strong learned association because of Model Approximation. Deployment Mediation may then fail to retrieve the current official record or may place it ineffectively in the context. During Autoregressive Realization, the system may select the former name and generate policy details consistent with that selection. The delivered response can consequently contain both the initial outdated claim and additional false claims produced through propagation.

The same interactions can instead correct an initial susceptibility. Retrieval can compensate for the model’s temporal boundary if the current record enters the operative context in usable form and influences generation. Decoding or verification may then prevent an unsupported continuation from reaching the user. Later production stages can therefore reduce, transmit, or amplify susceptibility arising at an earlier stage.

Different production paths can also produce the same observed error. An incorrect date might reflect a weak learned representation, an outdated retrieved source, selection among competing continuations, or

alteration during post-generation processing. The framework does not assign the error exclusively to one family. It identifies the combinations of conditions through which the error could have been produced or prevented.

This interaction changes the appropriate unit of analysis. A model name alone does not identify the information resources, context-construction process, decoding settings, tools, or post-generation controls used to produce a response. A reported hallucination rate therefore belongs to a configured system operating on a specified task, not to a model name alone. Controlled model-level comparisons remain useful when the other conditions are held constant.

G. Discussion and Conclusion

G.1 A response-production approach

The framework suggests a **response-production approach** to hallucination evaluation and management. Once a hallucination has been identified, the analysis should not automatically treat the trained model as its only possible source. It should describe the model, task, information resources, context-construction process, decoding configuration, tools, and relevant output controls. Management can then consider which parts of that production sequence may prevent, transmit, or amplify error.

This approach does not require a complete diagnosis of every individual response. Some controls can address several pathways. Retrieval can supply absent or outdated information. Concise response requirements can reduce cumulative exposure. Verification can detect errors arising at different production stages. But no control works equally well in every setting. Lower-temperature decoding cannot supply information that is absent, and retrieval cannot correct an error when the retrieved material is outdated, misleading, or ineffectively used.

The approach also directs attention to interactions among controls and system components. Expanding accessible evidence will not help when context construction excludes it. A reliable tool result will not help when the model incorporates it incorrectly. A stronger model may still answer a current-information question incorrectly when the deployed system supplies no current data. Improving one component can therefore have limited value when another remains inadequate.

G.2 Interpreting temporal hallucination more appropriately

Consider a model completed or released in February and deployed without access to current market data. If a user asks for Company X's stock price on May 31 and the system supplies an incorrect price, the claim is factually wrong and may be classified as hallucinated. But the result does not show that the trained model failed to learn or retain that fact. The requested price did not exist at the model's effective information boundary, and the deployment provided no means of obtaining it.

Classifying the delivered claim as hallucinated identifies the error; it does not establish that the error originated in model formation. An evaluation is therefore incomplete if it attributes the error solely to the trained model or compares systems without accounting for their access to current information. The result instead reveals a mismatch between the task and the system's deployment state. A task requiring current prices needs a current-data capability, such as a credible market-data service, a reliable tool connection, or a robust and well-governed retrieval process.

This interpretation does not excuse the wrong answer. A suitably designed system should recognize its information boundary, consult a current source, qualify the answer, or abstain rather than fabricate a price. The example instead changes what can reasonably be learned from the failure. It shows why the deployment state must remain adequate to the information demands of the task.

The same principle applies beyond finance. Law, officeholders, product specifications, organizational policies, medical guidance, and continuing-project state may all change after model formation. Periodic model updates can narrow some gaps, but static retraining cannot keep every system continuously aligned with every task. Deployment therefore requires continuing maintenance. Information sources, indexes, tools, permissions, context policies, and controls must remain suitable for the work the system is expected to perform.

G.3 Testable implications and existing evidence

The framework produces six testable hypotheses, summarized in Table 1. The first five concern particular production pathways. The sixth states the interaction proposition: the effect of an intervention at one locus may depend on conditions at another.

Hypothesis	Principal locus	Testable implication and existing evidence
H1—Temporal supplementation	Model Approximation × Deployment Mediation	For facts established after t_0 , supplying applicable current evidence should reduce hallucination relative to an otherwise equivalent unaugmented condition. Current-knowledge studies are consistent with this prediction ([4], [5]).
H2—Realization sensitivity	Autoregressive Realization	Holding the model, input, and context constant, changing decoding should change the frequency or composition of hallucinated claims. Factuality-oriented decoding provides supporting evidence ([11]).
H3—Length decomposition	Autoregressive Realization	Longer responses should increase response-level error through cumulative exposure; in some tasks, later assertions should also become less reliable through facts exhaustion ([12]).
H4—Propagation	Autoregressive Realization	Correcting an early erroneous commitment should reduce dependent errors later in the response. Snowballing experiments support this pathway ([13]).
H5—Context mediation	Deployment Mediation × Model/Realization	Holding evidence constant, changing its selection, position, compression, or labeling should affect whether the response conforms to it ([5], [15]).
H6—Interaction	Across all three loci	The effect of an intervention at one locus should depend on conditions at the others. This proposition requires systematic factorial testing.

Table 1. Testable implications mapped to the response-production framework.

Evidence for one hypothesis does not establish the entire framework. Similarly, improvement following an intervention supports the tested pathway only within the task and system configuration examined. This

allows evidence to accumulate across studies without treating one benchmark result as a universal explanation of hallucination.

G.4 Limitations

The three families are broad analytical categories, not an exhaustive taxonomy. New architectures and deployment designs may require them to be refined. Their boundaries can also overlap in structured ways. Evidence placement, for example, is a deployment decision. The model’s sensitivity to that placement reflects Model Approximation and Autoregressive Realization.

The framework does not estimate how frequently each family contributes to hallucination or how important each one is. Those quantities are likely to vary across tasks, models, and configurations and must be examined empirically. The framework also does not determine the applicable source of truth. It begins after a hallucination has been identified and asks where in response production it may have arisen.

G.5 Conclusion

Hallucination is observed in a delivered response, but that response is produced through a sequence of model, generation, and deployment processes. This paper organizes plausible sources of hallucination into Model Approximation, Autoregressive Realization, and Deployment Mediation. These families can operate separately, reinforce one another, or compensate for one another.

The resulting response-production approach discourages treating the trained model as the automatic source of every hallucination. It also supports more informative evaluation. An incorrect current stock price from a static, unaugmented model is a factual error, but it also reveals that the deployment state is inadequate for the task. Evaluation should therefore consider what information and capabilities the configured system had. Management should keep those capabilities aligned with the tasks that the system is expected to perform.

Existing studies provide different degrees of support for the hallucination pathways identified in the framework, including temporal information, predictive uncertainty, decoding, response length, propagation, retrieval, and context use. Future research, including empirical tests around the six hypotheses formulated in G.3, can help better understand these pathways and their roles in generating hallucination.

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